FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ

PHYSICS

CLASS XII

WAVE OPTICS

---

1. In Young's double slit experiment, the separation between the slits is halved and distance between the slits and screen is doubled. The fringe width is
   (a) unchanged (b) halved
   (c) doubled (d) quadrupled
2. Young's double slit experiment is carried out by using green, red and blue light, one colour at a time. The fringe widths recorded are β_G, β_R, and β_B respectively. Then,
   (a) $\beta_G > \beta_B > \beta_R$ (b) $\beta_B > \beta_G > \beta_R$
   (c) $\beta_R > \beta_B > \beta_G$ (d) $\beta_R > \beta_G > \beta_B$
3. Which one of the following statements is true?
   (a) both light and sound waves can travel in vacuum
   (b) both light and sound waves in air are transverse
   (c) the sound waves in air are longitudinal, while the light waves are transverse
   (d) both light and sound waves in air are longitudinal
4. The frequency of light wave in a material is 2×10^{14} Hz and wavelength is 5000 A^0. The refractive index of material will be
   (a) 1.50 (b) 3.00 (c) 1.33 (d) 1.40
5. Interference was observed in an interference chamber, when air was present. Now the chamber is evacuated and if the same light is used, a careful observer will see
   (a) no interference
   (b) interference with bright bands
   (c) interference with dark bands
   (d) interference, in which width of the fringe will be slightly increased
6. Light travels through a glass plate of thickness t and having refractive index μ. If c is the velocity of light in vacuum, the time taken by the light to travel this thickness of glass is
   (a) tμc (b) tc/μ (c) μc/t (d) μt/c
7. A parallel beam of monochromatic light of wavelength 5000 A^o is incident normally on a single narrow slit of width 0.001 mm. The light is focused by a convex lens on a screen placed in focal plane. The first minimum will be formed for the angle of diffraction equal to
   (a) 0° (b) 15° (c) 30° (d) 50°
8. A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct?
   (a) diffraction pattern is not observed on the screen in the case of electrons
   (b) the angular width of the central maximum of the diffraction pattern will increase

FREEDOM INTERNATIONAL SCHOOL

(c) the angular width of the central maximum will decrease
(d) the angular width of the central maximum will be unaffected

9. At the first minimum adjacent to the central maximum of a single slit diffraction pattern the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is

(a) $\pi/8$ radian (b) $\pi/4$ radian (c) $\pi/2$ radian (d) π radian

10. A beam of light of $\lambda = 600$ nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is

(a) 1.2 cm (b) 1.2 mm (c) 2.4 cm (d) 2.4 mm

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** A thin film such as soap bubble or a thin layer of oil on water show beautiful colours when illuminated by white light.
Reason: It happens due to the interference of light reflected from upper and lower faces of the thin film.
12. **Assertion:** Diffraction takes place for all types of waves: mechanical or non-mechanical, transverse or longitudinal.
Reason: Diffraction effects are perceptible only if the wavelength of the wave is comparable to the dimensions of the diffracting device.
13. **Assertion:** Interference is not observed if the two coherent slit sources are broad.
Reason: A broad source is equivalent to many narrow slit sources.
14. **Assertion:** For best contrast between maxima and minima in the interference pattern of Young's double slit experiment, the amplitudes of light waves emerging from the two sources should be equal.
Reason: For interference, the sources must be coherent.
15. **Assertion:** When a thin transparent sheet is placed in front of both the slits of Young's experiment, the fringe width will remain the same.
Reason: In Young's experiment, the fringe width is directly proportional to the wavelength of the source used.

FREEDOM INTERNATIONAL SCHOOL